SN74LVC1G86 Single 2-Input Exclusive-OR Gate

1 Features

- Qualified from –40°C to +125°C
- Supports 5V V_{CC} operation
- Inputs are over voltage tolerant up to 5.5V
- Supports down translation to V_{CC}
- Maximum t_{pd} of 4ns at 3.3V and 15pF load
- Low power consumption, 10µA maximum I_{CC} at 85°C
- ±24mA output drive at 3.3V
- I_{off} supports partial-power-down mode, and back-drive protection
- Available in the Texas Instruments NanoFree™ package
- Latch-up performance exceeds 100mA per JESD 78, Class II

2 Applications

- Wireless headsets
- Motor drives and controls
- TVs
- Set-top boxes
- Audio

3 Description

The SN74LVC1G86 device performs the Boolean function $Y = \overline{A}B + A\overline{B}$ in positive logic. This single 2-input exclusive-OR gate is designed for operation from 1.65V to 5.5V V_{CC}.

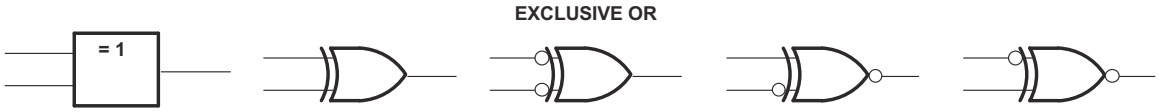
If the input is low, the other input is reproduced in true form at the output. If the input is high, the signal on the other input is reproduced inverted at the output. This device has low power consumption with maximum t_{pd} of 4ns at 3.3V and 15pF capacitive load. The maximum output drive is ±32mA at 4.5V and ±24mA at 3.3V.

This device is fully specified for partial-power-down applications using I_{off}. The I_{off} circuitry disables the outputs, preventing damaging current back flow through the device when it is powered down.

Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
SN74LVC1G86DBV	SOT-23 (5)	2.90mm × 1.60mm
SN74LVC1G86DCK	SC70 (5)	2.00mm × 1.25mm
SN74LVC1G86DRL	SOT (5)	1.60mm × 1.20mm
SN74LVC1G86YZP	DSBGA (5)	1.44mm × 0.94mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Copyright © 2017, Texas Instruments Incorporated

Functional Block Diagram

Table of Contents

1 Features....1
2 Applications....1
3 Description....1
4 Pin Configuration and Functions....3
5 Specifications....4
5.1 Absolute Maximum Ratings....4
5.2 ESD Ratings....4
5.3 Recommended Operating Conditions....4
5.4 Thermal Information....5
5.5 Electrical Characteristics....5
5.6 Switching Characteristics, C_L = 15pF....6
5.7 Switching Characteristics, C_L = 30pF or 50pF....6
5.8 Operating Characteristics....6
5.9 Typical Characteristics....6
6 Parameter Measurement Information....7
7 Detailed Description....8
7.1 Overview....8
7.2 Functional Block Diagram....8
7.3 Feature Description....8
7.4 Function Table....9
8 Application and Implementation....10
8.1 Application Information....10
8.2 Typical Application....10
8.3 Power Supply Recommendations....11
8.4 Layout....11
9 Device and Documentation Support....13
9.1 Receiving Notification of Documentation Updates....13
9.2 Support Resources....13
9.3 Trademarks....13
9.4 Electrostatic Discharge Caution....13
9.5 Glossary....13
10 Revision History....13
11 Mechanical, Packaging, and Orderable Information....13

Copyright © 2026 Texas Instruments Incorporated

4 Pin Configuration and Functions

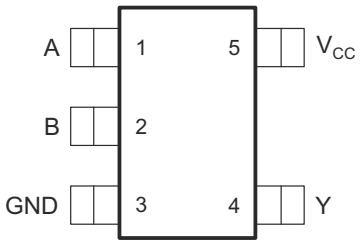


Figure 4-1. DBV Package 5-Pin SOT-23 Top View

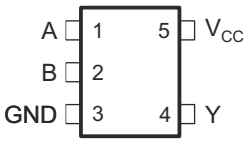


Figure 4-2. DRL Package 5-Pin SOT Top View

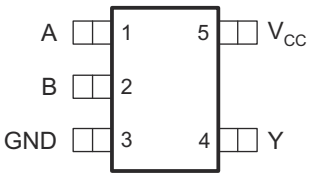


Figure 4-3. DCK Package 5-Pin SC70 Top View

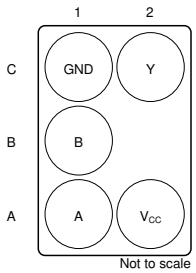


Figure 4-4. YZP Package 5-Pin DSBGA Bottom View

Pin Functions

PIN			TYPE[1]	DESCRIPTION
NAME	DBV, DRL, DCK	DSBGA		
A	1	A1	I	Input A
B	2	B1	I	Input B
GND	3	C1	—	Ground
V_{CC}	5	A2	—	Positive Supply
Y	4	C2	O	Output Y

(1) I = input, O = output

Copyright © 2026 Texas Instruments Incorporated

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage		–0.5	6.5	V
V_I	Input voltage(2)		–0.5	6.5	V
V_O	Voltage applied to any output in the high-impedance or power-off state(2)		–0.5	6.5	V
V_O	Voltage applied to any output in the high or low state(2) (3)		–0.5	V_{CC} + 0.5	V
I_{IK}	Input clamp current	V_I < 0		–50	mA
I_{OK}	Output clamp current	V_O < 0		–50	mA
I_O	Continuous output current			±50	mA
	Continuous current through V_{CC} or GND			±100	mA
T_J	Junction temperature			150	°C
T_{stg}	Storage temperature		–65	150	°C

(1) Operation outside the *Absolute Maximum Ratings* may cause permanent device damage. *Absolute Maximum Ratings* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Conditions*. If used outside the *Recommended Operating Conditions* but within the *Absolute Maximum Ratings*, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) The input negative-voltage and output voltage ratings can be exceeded if the input and output clamp-current ratings are observed.
(3) The value of V_{CC} is provided in the *Recommended Operating Conditions* table.

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	1000	

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage	Operating	1.65	5.5	V
		Data retention only	1.5		
V_{IH}	High-level input voltage	V_{CC} = 1.65V to 1.95V	0.65 × V_{CC}		V
		V_{CC} = 2.3V to 2.7V	1.7		
		V_{CC} = 3V to 3.6V	2		
		V_{CC} = 4.5V to 5.5V	0.7 × V_{CC}		
V_{IL}	Low-level input voltage	V_{CC} = 1.65V to 1.95V		0.35 × V_{CC}	V
		V_{CC} = 2.3V to 2.7V		0.7	
		V_{CC} = 3V to 3.6V		0.8	
		V_{CC} = 4.5V to 5.5V		0.3 × V_{CC}	
V_I	Input voltage		0	5.5	V
V_O	Output voltage		0	V_{CC}	V
I_{OH}	High-level output current	V_{CC} = 1.65V		–4	mA
		V_{CC} = 2.3V		–8	
		V_{CC} = 3V		–16	
				–24	
		V_{CC} = 4.5V		–32	

Copyright © 2026 Texas Instruments Incorporated

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
I_{OL}	Low-level output current	V_{CC} = 1.65V		4	mA
		V_{CC} = 2.3V		8	
		V_{CC} = 3V		16	
				24	
		V_{CC} = 4.5V		32	
$\Delta t/\Delta v$	Input transition rise or fall rate	V_{CC} = 1.8V ± 0.15V, 2.5V ± 0.2V		20	ns/V
		V_{CC} = 3.3V ± 0.3V		10	
		V_{CC} = 5V ± 0.5V		5	
T_A	Operating free-air temperature	YZP package	–40	85	°C
		DCK, DBV, and DRL packages	–40	125	

(1) All unused inputs of the device must be held at V_{CC} or GND to maintain proper device operation. Refer to the TI application note, *Implications of Slow or Floating CMOS Inputs*.

5.4 Thermal Information

PACKAGE	PINS	THERMAL METRIC(1)						UNIT
		$R_{\theta JA}$	$R_{\theta JC(top)}$	$R_{\theta JB}$	Ψ_{JT}	Ψ_{JB}	$R_{\theta JC(bot)}$	
DBV (SOT-23)	5	357.1	263.7	264.4	195.6	262.2	-	°C/W
DCK (SOT-SC70)	5	371	297.5	258.6	195.6	256.2	-	°C/W
YZP (DSBGA)	5	130	-	-	-	-	-	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application note.

5.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS			V_{CC}	MIN	TYP(1)	MAX	UNIT
V_{OH}	I_{OH} = –100 µA			1.65V to 5.5V	V_{CC} – 0.1			V
	I_{OH} = –4mA			1.65V	1.2			
	I_{OH} = –8mA			2.3V	1.9			
	I_{OH} = –16mA			3V	2.4			
	I_{OH} = –24mA				2.3			
	I_{OH} = –32mA			4.5V	3.8			
V_{OL}	I_{OL} = 100 µA			1.65V to 5.5V			0.1	V
	I_{OL} = 4mA			1.65V			0.45	
	I_{OL} = 8mA			2.3V			0.3	
	I_{OL} = 16mA			3V			0.4	
	I_{OL} = 24mA						0.55	
	I_{OL} = 32mA			4.5V			0.55	
I_I A or B input	V_I = 5.5V or GND			0 to 5.5V			±5	µA
I_{off}	V_I or V_O = 5.5V			0			±10	µA
I_{CC}	V_I = V_{CC} or GND,	I_O = 0	–40°C to 85°C	1.65V to 5.5V			10	µA
			–40°C to 125°C				15	
ΔI_{CC}	One input at V_{CC} – 0.6V, Other inputs at V_{CC} or GND			3V to 5.5V			500	µA
C_i	V_I = V_{CC} or GND			3.3V		6		pF

(1) All typical values are at V_{CC} = 3.3V, T_A = 25°C.

Copyright © 2026 Texas Instruments Incorporated

5.6 Switching Characteristics, C_L = 15pF

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 6-1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	V_{CC} = 1.8V ± 0.15V		V_{CC} = 2.5V ± 0.2V		V_{CC} = 3.3V ± 0.3V		V_{CC} = 5V ± 0.5V		UNIT
				MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t_{pd}	A or B	Y	T_A = –40°C to +85°C	2.1	9.1	1	4.9	0.6	4	0.8	3.3	ns

5.7 Switching Characteristics, C_L = 30pF or 50pF

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	V_{CC} = 1.8V ± 0.15V		V_{CC} = 2.5V ± 0.2V		V_{CC} = 3.3V ± 0.3V		V_{CC} = 5V ± 0.5V		UNIT
				MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t_{pd}	A or B	Y	–40°C to +85°C temperature range, see Figure 6-1	3.5	9.9	1.8	5.5	1.3	5	1	4	ns
			–40°C to +125°C temperature range, see Figure 6-1	3.5	12	1.8	7	1.3	6	1	5	

5.8 Operating Characteristics

T_A = 25°C

PARAMETER		TEST CONDITIONS	V_{CC} = 1.8V	V_{CC} = 2.5V	V_{CC} = 3.3V	V_{CC} = 5V	UNIT
			TYP	TYP	TYP	TYP	
C_{pd}	Power dissipation capacitance	f = 10MHz	22	22	22	24	pF

5.9 Typical Characteristics

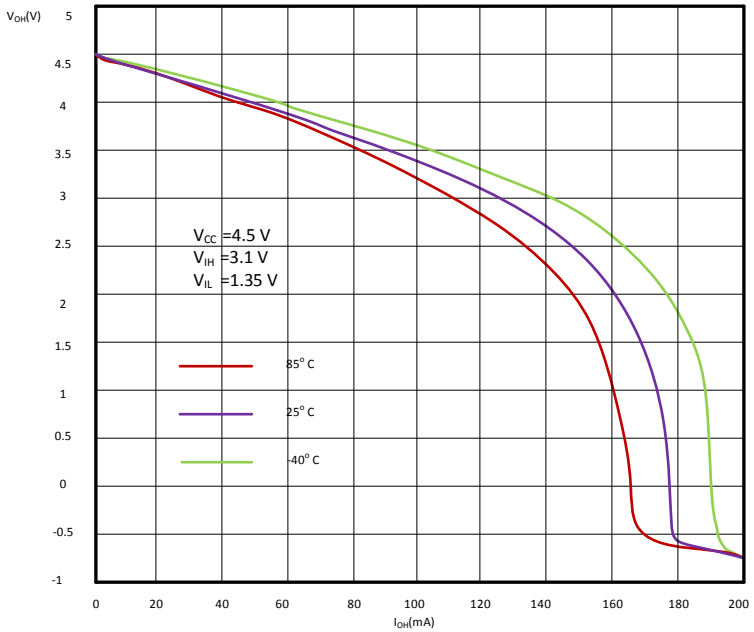
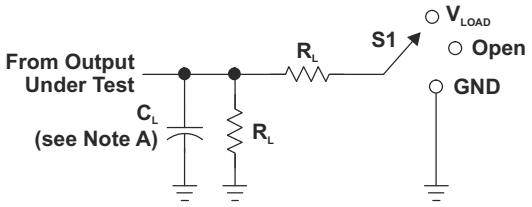


Figure 5-1. V_{oh} vs I_{oh} at 4.5V

Copyright © 2026 Texas Instruments Incorporated

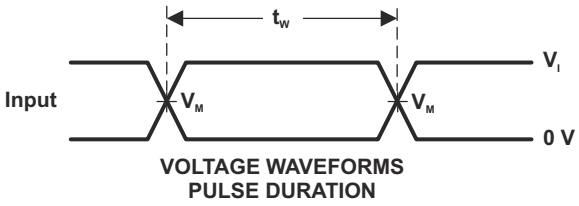
6 Parameter Measurement Information



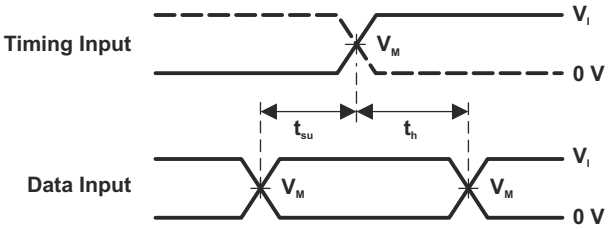
LOAD CIRCUIT

TEST	S1
t_{PLH}/t_{PHL}	Open
t_{PLZ}/t_{PZL}	V_{LOAD}
t_{PHZ}/t_{PZH}	GND

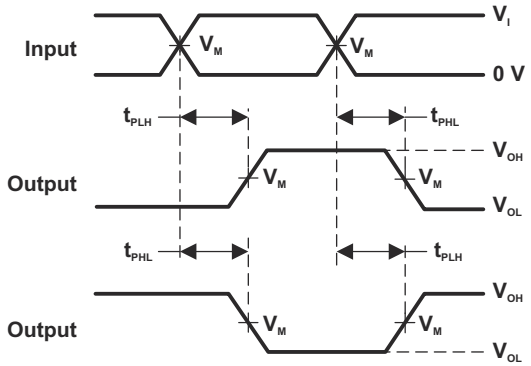
V_{CC}	INPUTS		V_M	V_{LOAD}	C_L	R_L	V_D
	V_I	t_r/t_f					
1.8 V ± 0.15 V	V_{CC}	≤2 ns	$V_{CC}/2$	2 × V_{CC}	15 pF	1 MΩ	0.15 V
2.5 V ± 0.2 V	V_{CC}	≤2 ns	$V_{CC}/2$	2 × V_{CC}	15 pF	1 MΩ	0.15 V
3.3 V ± 0.3 V	3 V	≤2.5 ns	1.5 V	6 V	15 pF	1 MΩ	0.3 V
5 V ± 0.5 V	V_{CC}	≤2.5 ns	$V_{CC}/2$	2 × V_{CC}	15 pF	1 MΩ	0.3 V



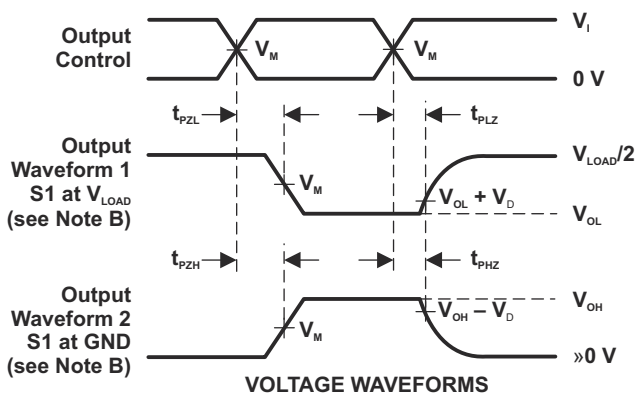
VOLTAGE WAVEFORMS
PULSE DURATION



VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES
INVERTING AND NONINVERTING OUTPUTS



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES
LOW- AND HIGH-LEVEL ENABLING

NOTES: A. C_L includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, Z_O = 50 Ω.
D. The outputs are measured one at a time, with one transition per measurement.
E. t_{PLZ} and t_{PHZ} are the same as t_{dis}.
F. t_{PZL} and t_{PZH} are the same as t_{en}.
G. t_{PLH} and t_{PHL} are the same as t_{pd}.
H. All parameters and waveforms are not applicable to all devices.

Figure 6-1. Load Circuit and Voltage Waveforms

Copyright © 2026 Texas Instruments Incorporated

7 Detailed Description

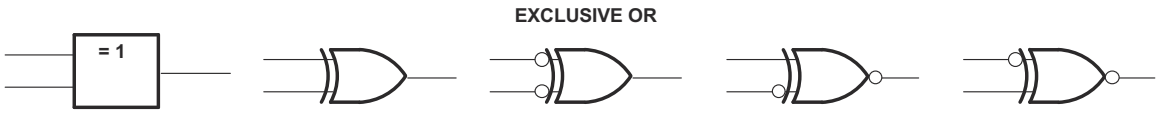
7.1 Overview

The SN74LVC1G86 device performs the Boolean function Y = $\overline{A}$B + A $\overline{B}$ in positive logic. This single 2-input exclusive-OR gate is designed for 1.65V to 5.5V V_{CC} operation.

A common application is as a true and complement element. If the input is low, the other input is reproduced in true form at the output. If the input is high, the signal on the other input is reproduced inverted at the output.

NanoFree package technology is a major breakthrough in IC packaging concepts, using the die as the package. This device is fully specified for partial-power-down applications using I_{off}. The I_{off} circuitry disables the outputs, preventing damaging current backflow through the device when powered down.

7.2 Functional Block Diagram



Copyright © 2017, Texas Instruments Incorporated

An exclusive-OR gate has many applications, some of which can be represented better by alternative logic symbols. These are five equivalent exclusive-OR symbols valid for an SN74LVC1G86 gate in positive logic; negation can be shown at any two ports.

7.3 Feature Description

7.3.1 Balanced High-Drive CMOS Push-Pull Outputs

A balanced output allows the device to sink and source similar currents. The high drive capability of this device creates fast edges into light loads so routing and load conditions must be considered to prevent ringing. Additionally, the outputs of this device are capable of driving larger currents than the device can sustain without being damaged. Limit the power output of the device to avoid thermal runaway and damage due to over-current. The electrical and thermal limits defined the in the *Section 5.1* must be followed at all times.

7.3.2 Standard CMOS Inputs

Standard CMOS inputs are high impedance and are typically modeled as a resistor in parallel with the input capacitance given in the *Section 5.5*. The worst case resistance is calculated with the maximum input voltage, given in the *Section 5.3*, and the maximum input leakage current, given in the *Section 5.5*, using Ohm's law (R = V ÷ I).

Signals applied to the inputs need to have fast edge rates, as defined by Δt/Δv in *Section 5.3* to avoid excessive currents and oscillations. If tolerance to a slow or noisy input signal is required, a device with a Schmitt-trigger input must be used to condition the input signal prior to the standard CMOS input.

Copyright © 2026 Texas Instruments Incorporated

7.3.3 Clamp Diodes

The inputs and outputs to this device have negative clamping diodes.

CAUTION

Avoid any voltage below or above the input or output voltage specified in the *Section 5.1*. In this event, the current must be limited to the maximum input or output clamp current value indicated in the *Section 5.1* to avoid damage to the device.

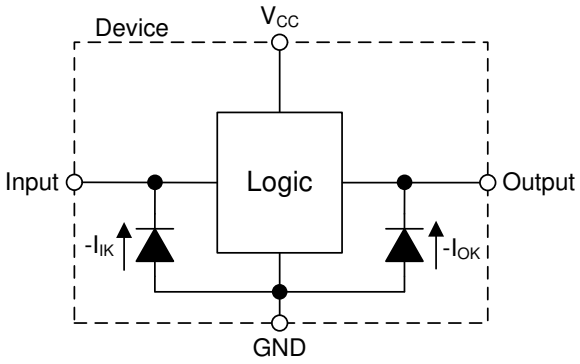


Figure 7-1. Electrical Placement of Clamping Diodes for Each Input and Output

7.3.4 Partial Power Down (I_{off})

The inputs and outputs for this device enter a high impedance state when the supply voltage is 0V. The maximum leakage into or out of any input or output pin on the device is specified by I_{off} in the *Section 5.5*.

7.3.5 Over-voltage Tolerant Inputs

Input signals to this device can be driven above the supply voltage so long as they remain below the maximum input voltage value specified in the *Section 5.3*.

7.4 Function Table

Table 7-1 lists the functional modes of the SN74LVC1G86 device.

Table 7-1. Function Table

INPUTS		OUTPUT
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

The SN74LVC1G86 device can accept input voltages up to 5.5V at any valid V_{CC} which makes the device an excellent choice for down translation. Therefore, the SN74LVC1G86 is designed for various bus interface applications.

8.2 Typical Application

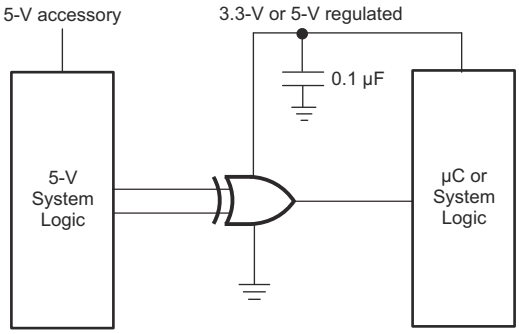


Figure 8-1. Typical Application Schematic

8.2.1 Design Requirements

This device uses CMOS technology and has balanced output drive. Avoid bus contention because it can drive currents that exceed maximum limits. The high drive also creates fast edges into light loads, so routing and load conditions must be considered to prevent ringing.

8.2.2 Detailed Design Procedure

1. Recommended Input Conditions
   - For rise time and fall time specifications, see Δt/ΔV in the *Section 5.3* table.
   - For specified High and low levels, see V_{IH} and V_{IL} in the *Section 5.3* table.
   - Inputs are overvoltage tolerant allowing them to go as high as 5.5V at any valid V_{CC}.
2. Recommended Output Conditions
   - Load currents must not exceed 32mA per output and 50mA total for the part.
   - Outputs must not be pulled above V_{CC}.

Copyright © 2026 Texas Instruments Incorporated

8.2.3 Application Curve

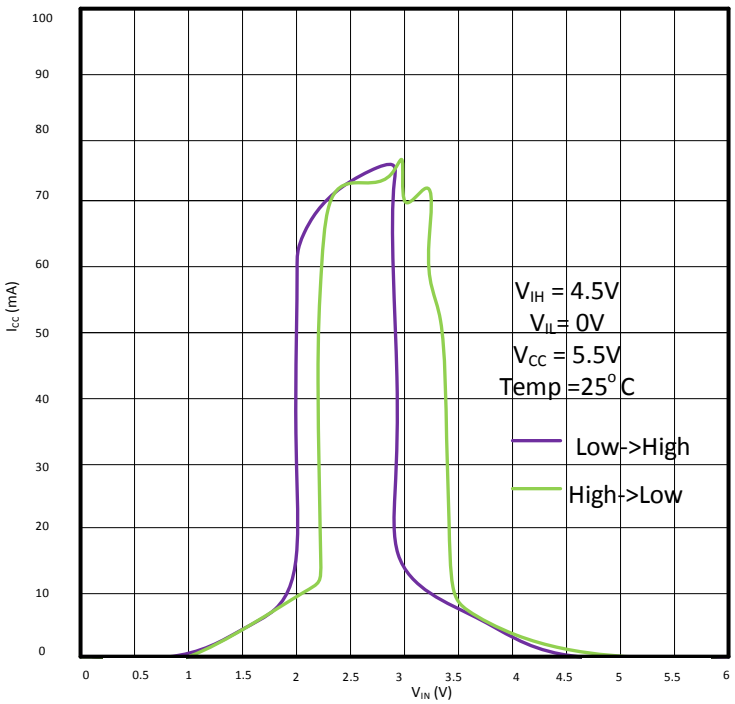


Figure 8-2. I_{CC} vs. V_{IN}

8.3 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Section 5.3* table.

Each V_{CC} pin must have a good bypass capacitor to prevent power disturbance. For devices with a single supply, 0.1µF is recommended. If there are multiple V_{CC} pins, 0.01µF or 0.022µF is recommended for each power pin. Parallel multiple bypass caps to reject different frequencies of noise. A 0.1µF and 1µF are commonly used in parallel. The bypass capacitor must be installed as close to the power pin as possible for best results.

8.4 Layout

8.4.1 Layout Guidelines

Even low data rate digital signals can have high frequency signal components due to fast edge rates. When a PCB trace turns a corner at a 90° angle, a reflection can occur. A reflection occurs primarily because of the change of width of the trace. At the apex of the turn, the trace width increases to 1.414 times the width. This increase upsets the transmission-line characteristics, especially the distributed capacitance and self–inductance of the trace which results in the reflection. Not all PCB traces can be straight and therefore some traces must turn corners. Figure 8-3 shows progressively better techniques of rounding corners. Only the last example (BEST) maintains constant trace width and minimizes reflections.

Copyright © 2026 Texas Instruments Incorporated

8.4.2 Layout Example

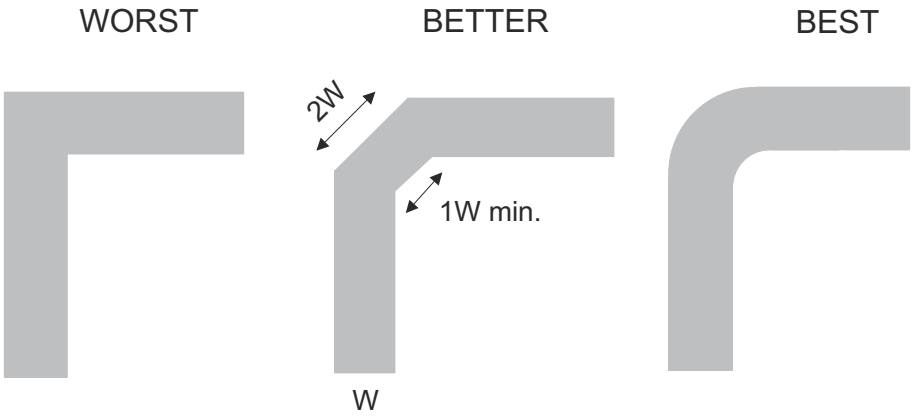


Figure 8-3. Trace Example

Copyright © 2026 Texas Instruments Incorporated

9 Device and Documentation Support

9.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

9.2 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

9.3 Trademarks

NanoFree™ and TI E2E™ are trademarks of Texas Instruments.
All trademarks are the property of their respective owners.

9.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.5 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from June 1, 2017 to August 12, 2026 (from Revision Q (June 2017) to Revision R (August 2026)) — **Page**

- Added *Thermal Information* table.... 5
- Changed Junction-to-ambient thermal resistance value for DBV package from: 206°C/W to: 357.1°C/W.... 5
- Changed Junction-to-ambient thermal resistance value for DCK package from: 252°C/W to: 371°C/W.... 5
- Changed tpd maximum value for V_{CC} = 2.5V ± 0.2V from: 4.5ns to: 4.9ns.... 6

Changes from Revision P (September 2015) to Revision Q (June 2017) — **Page**

- Changed YZP (DSBGA) package pinout diagram and added DSBGA column.... 3
- Added *Balanced High-Drive CMOS Push-Pull Outputs*, *Standard CMOS Inputs*, *Clamp Diodes*, *Partial Power Down (I_{off})*, and *Over-voltage Tolerant Inputs* sections.... 8

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

Copyright © 2026 Texas Instruments Incorporated

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74LVC1G86DBVR	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 125	(C865, C86F, C86J, C86K, C86R)
SN74LVC1G86DBVR.A	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(C865, C86F, C86J, C86K, C86R)
SN74LVC1G86DBVR.B	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(C865, C86F, C86J, C86K, C86R)
SN74LVC1G86DBVRE4	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C86F
SN74LVC1G86DBVRG4	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C86F
SN74LVC1G86DBVRG4.B	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C86F
SN74LVC1G86DBVT	Active	Production	SOT-23 (DBV) | 5	250 | SMALL T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 125	(C865, C86F, C86J, C86K, C86R)
SN74LVC1G86DBVT.B	Active	Production	SOT-23 (DBV) | 5	250 | SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(C865, C86F, C86J, C86K, C86R)
SN74LVC1G86DCKR	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 125	(CH5, CHF, CHJ, CHK, CHR)
SN74LVC1G86DCKR.A	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(CH5, CHF, CHJ, CHK, CHR)
SN74LVC1G86DCKR.B	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(CH5, CHF, CHJ, CHK, CHR)
SN74LVC1G86DCKRE4	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	CH5
SN74LVC1G86DCKRG4	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	CH5
SN74LVC1G86DCKRG4.B	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	CH5
SN74LVC1G86DCKT	Active	Production	SC70 (DCK) | 5	250 | SMALL T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 125	(CH5, CHF, CHJ, CHK, CHR)
SN74LVC1G86DCKT.B	Active	Production	SC70 (DCK) | 5	250 | SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(CH5, CHF, CHJ, CHK, CHR)
SN74LVC1G86DCKTG4	Active	Production	SC70 (DCK) | 5	250 | SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	CH5
SN74LVC1G86DCKTG4.B	Active	Production	SC70 (DCK) | 5	250 | SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	CH5
SN74LVC1G86DRLR	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	(CH7, CHR)
SN74LVC1G86DRLR.A	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	(CH7, CHR)
SN74LVC1G86DRLR.B	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	(CH7, CHR)
SN74LVC1G86YZPR	Active	Production	DSBGA (YZP) | 5	3000 | LARGE T&R	Yes	SNAGCU	Level-1-260C-UNLIM	-40 to 85	(CH7, CHN)

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74LVC1G86YZPR.B	Active	Production	DSBGA (YZP) | 5	3000 | LARGE T&R	Yes	SNAGCU	Level-1-260C-UNLIM	-40 to 85	(CH7, CHN)

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

OTHER QUALIFIED VERSIONS OF SN74LVC1G86 :

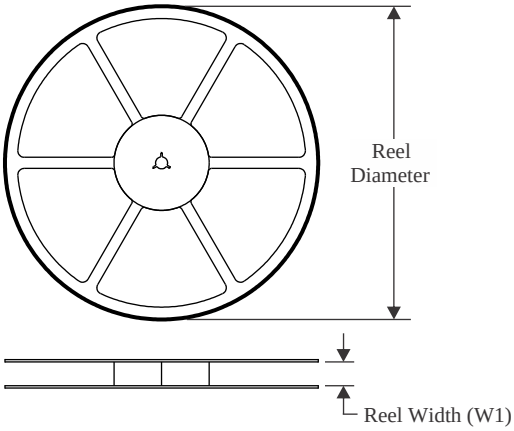
- Automotive : SN74LVC1G86-Q1
- Enhanced Product : SN74LVC1G86-EP

NOTE: Qualified Version Definitions:

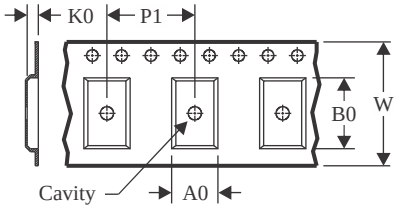
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects
- Enhanced Product - Supports Defense, Aerospace and Medical Applications

TAPE AND REEL INFORMATION

REEL DIMENSIONS

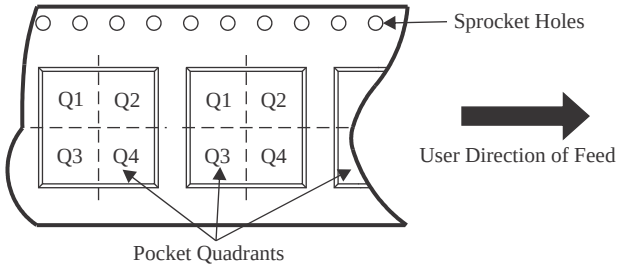


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE

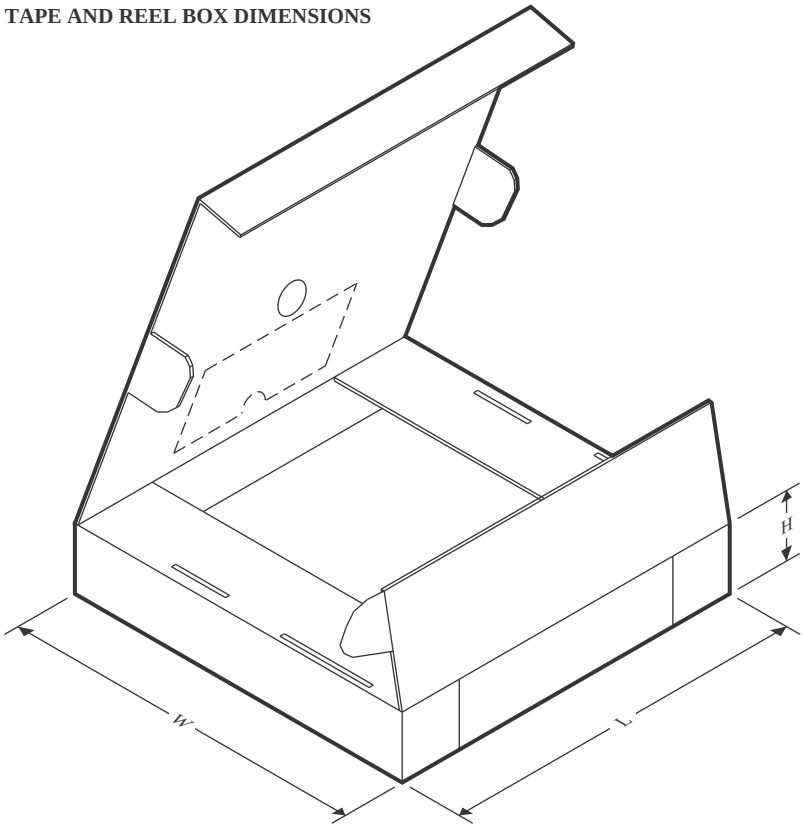


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	178.0	9.2	3.3	3.23	1.55	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	178.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	178.0	9.0	3.3	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVRG4	SOT-23	DBV	5	3000	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G86DBVT	SOT-23	DBV	5	250	178.0	9.0	3.3	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVT	SOT-23	DBV	5	250	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DBVT	SOT-23	DBV	5	250	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G86DBVT	SOT-23	DBV	5	250	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G86DBVT	SOT-23	DBV	5	250	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G86DCKR	SC70	DCK	5	3000	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G86DCKR	SC70	DCK	5	3000	180.0	8.4	2.3	2.55	1.2	4.0	8.0	Q3
SN74LVC1G86DCKR	SC70	DCK	5	3000	180.0	8.4	2.47	2.3	1.25	4.0	8.0	Q3

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74LVC1G86DCKR	SC70	DCK	5	3000	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G86DCKR	SC70	DCK	5	3000	178.0	8.4	2.25	2.45	1.2	4.0	8.0	Q3
SN74LVC1G86DCKR	SC70	DCK	5	3000	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G86DCKRG4	SC70	DCK	5	3000	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G86DCKT	SC70	DCK	5	250	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G86DCKT	SC70	DCK	5	250	180.0	8.4	2.47	2.3	1.25	4.0	8.0	Q3
SN74LVC1G86DCKT	SC70	DCK	5	250	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G86DCKT	SC70	DCK	5	250	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G86DCKTG4	SC70	DCK	5	250	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G86DRLR	SOT-5X3	DRL	5	4000	180.0	8.4	1.98	1.78	0.69	4.0	8.0	Q3
SN74LVC1G86YZPR	DSBGA	YZP	5	3000	178.0	9.2	1.02	1.52	0.63	4.0	8.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	208.0	191.0	35.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	210.0	185.0	35.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	202.0	201.0	28.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G86DBVR	SOT-23	DBV	5	3000	210.0	185.0	35.0
SN74LVC1G86DBVRG4	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G86DBVT	SOT-23	DBV	5	250	180.0	180.0	18.0
SN74LVC1G86DBVT	SOT-23	DBV	5	250	210.0	185.0	35.0
SN74LVC1G86DBVT	SOT-23	DBV	5	250	202.0	201.0	28.0
SN74LVC1G86DBVT	SOT-23	DBV	5	250	180.0	180.0	18.0
SN74LVC1G86DBVT	SOT-23	DBV	5	250	210.0	185.0	35.0
SN74LVC1G86DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G86DCKR	SC70	DCK	5	3000	210.0	185.0	35.0
SN74LVC1G86DCKR	SC70	DCK	5	3000	202.0	201.0	28.0
SN74LVC1G86DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G86DCKR	SC70	DCK	5	3000	208.0	191.0	35.0

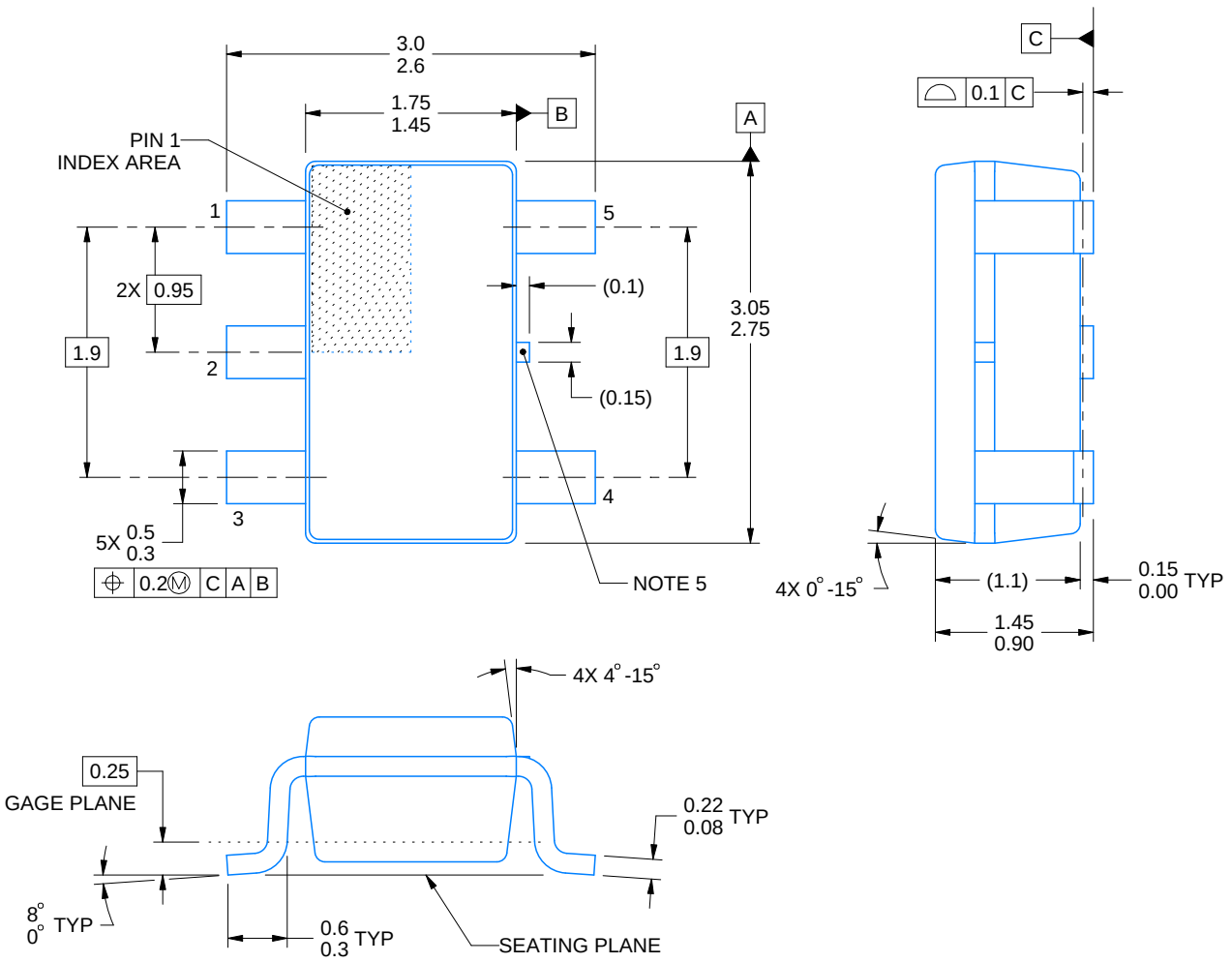
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74LVC1G86DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G86DCKRG4	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G86DCKT	SC70	DCK	5	250	180.0	180.0	18.0
SN74LVC1G86DCKT	SC70	DCK	5	250	202.0	201.0	28.0
SN74LVC1G86DCKT	SC70	DCK	5	250	180.0	180.0	18.0
SN74LVC1G86DCKT	SC70	DCK	5	250	180.0	180.0	18.0
SN74LVC1G86DCKTG4	SC70	DCK	5	250	180.0	180.0	18.0
SN74LVC1G86DRLR	SOT-5X3	DRL	5	4000	202.0	201.0	28.0
SN74LVC1G86YZPR	DSBGA	YZP	5	3000	220.0	220.0	35.0

PACKAGE OUTLINE

DBV0005A

SOT-23 - 1.45 mm max height

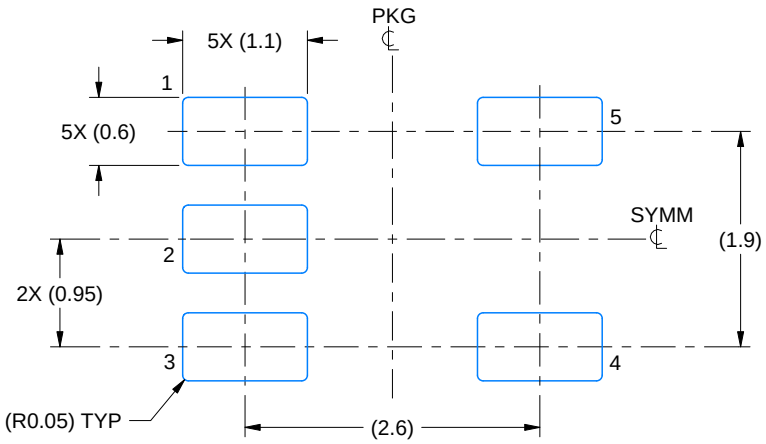
SMALL OUTLINE TRANSISTOR



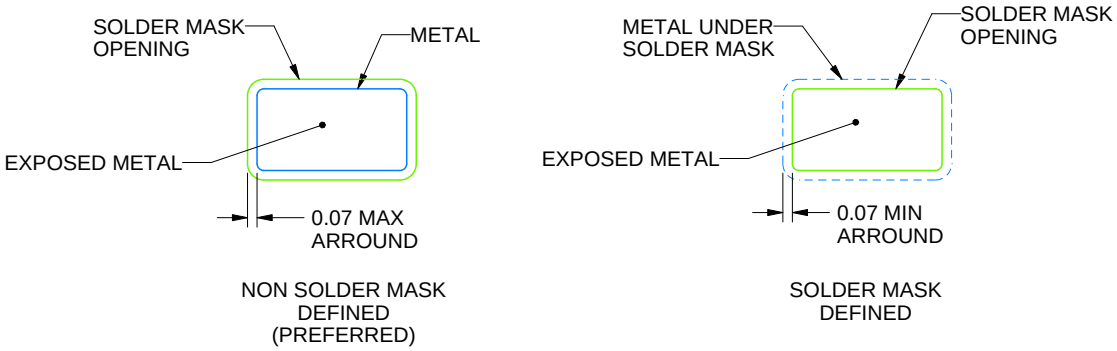
4214839/K 08/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Refernce JEDEC MO-178.
4. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25 mm per side.
5. Support pin may differ or may not be present.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X

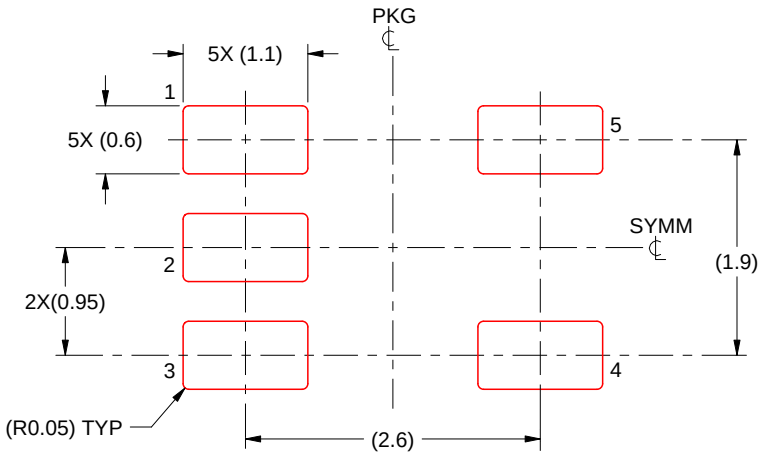


SOLDER MASK DETAILS

4214839/K 08/2024

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

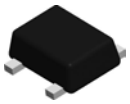


SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214839/K 08/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

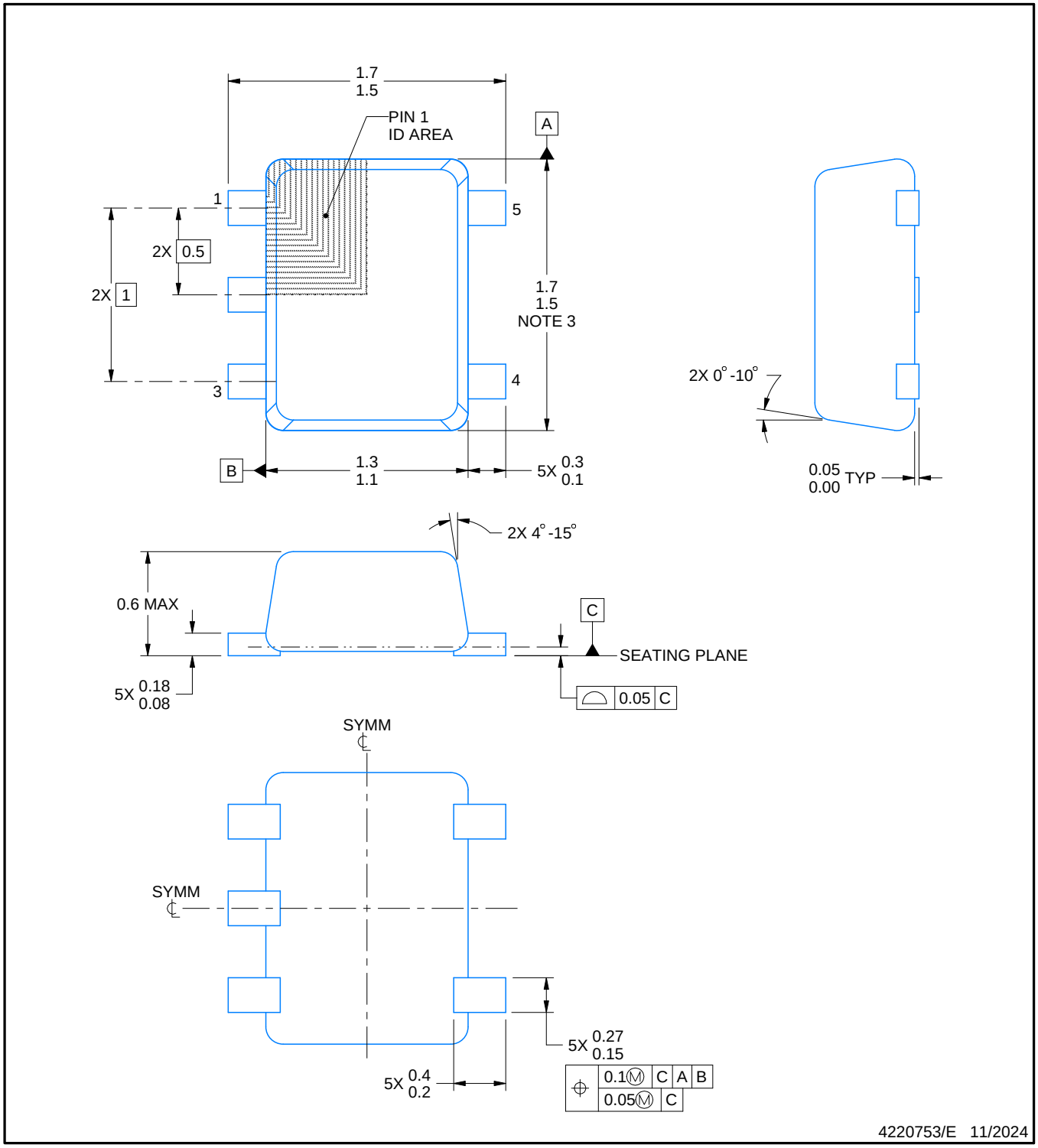


PACKAGE OUTLINE

DRL0005A

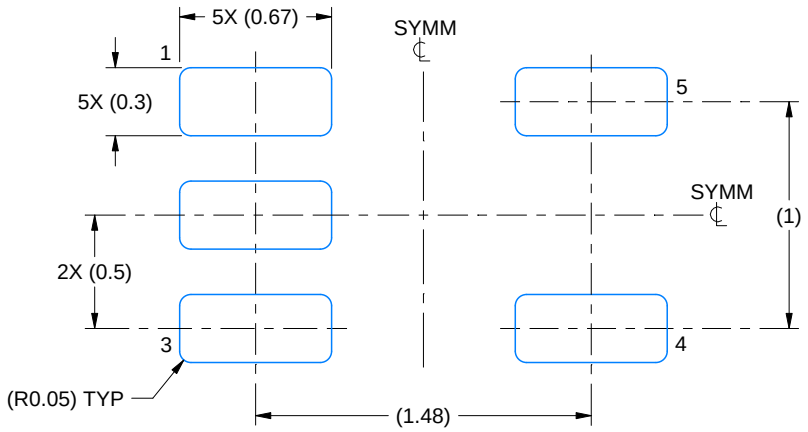
SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE

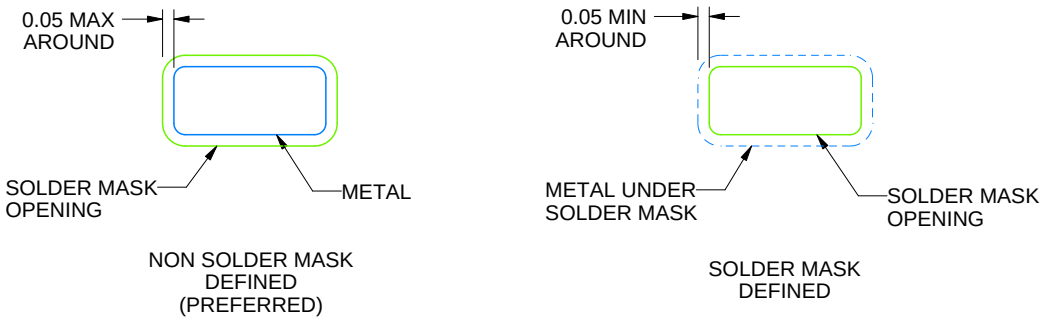


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-293 Variation UAAD-1



LAND PATTERN EXAMPLE
SCALE:30X

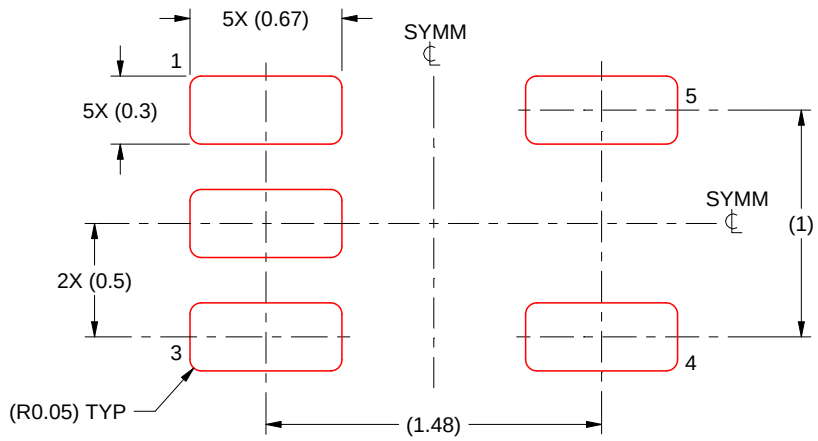


SOLDERMASK DETAILS

4220753/E 11/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:30X

4220753/E 11/2024

NOTES: (continued)

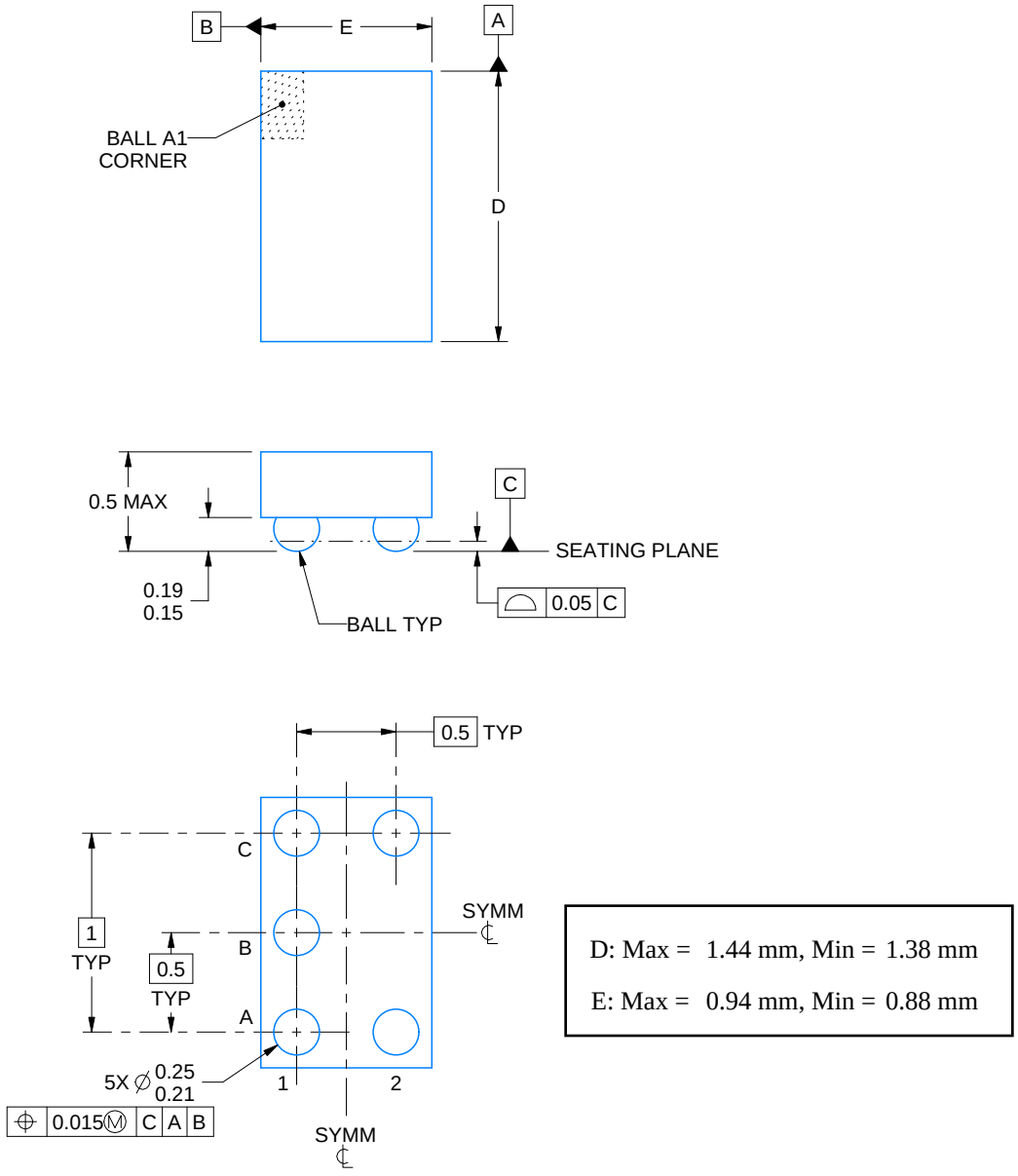
7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

PACKAGE OUTLINE

YZP0005

DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY



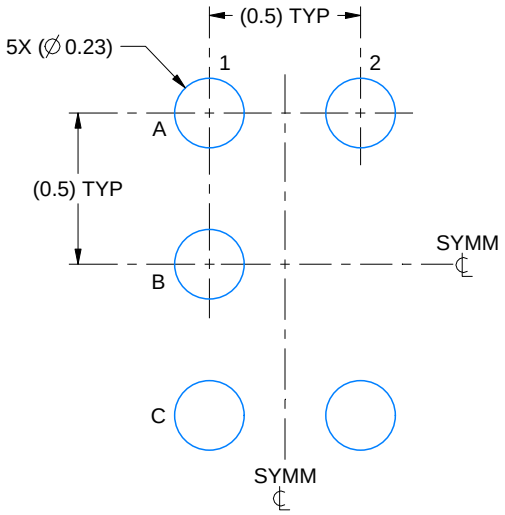
D: Max = 1.44 mm, Min = 1.38 mm

E: Max = 0.94 mm, Min = 0.88 mm

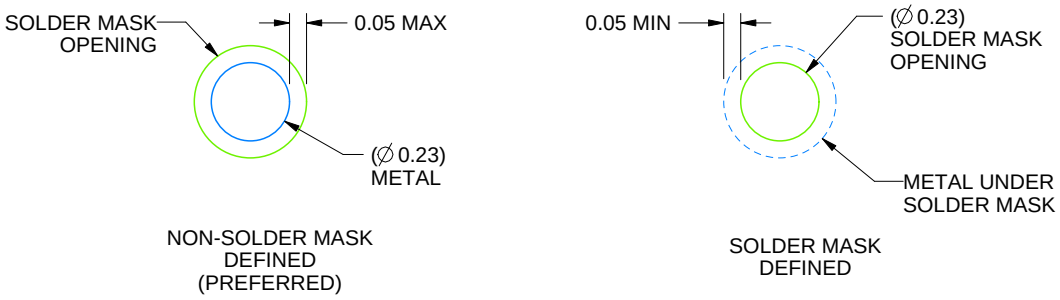
4219492/A 05/2017

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.



LAND PATTERN EXAMPLE
SCALE:40X

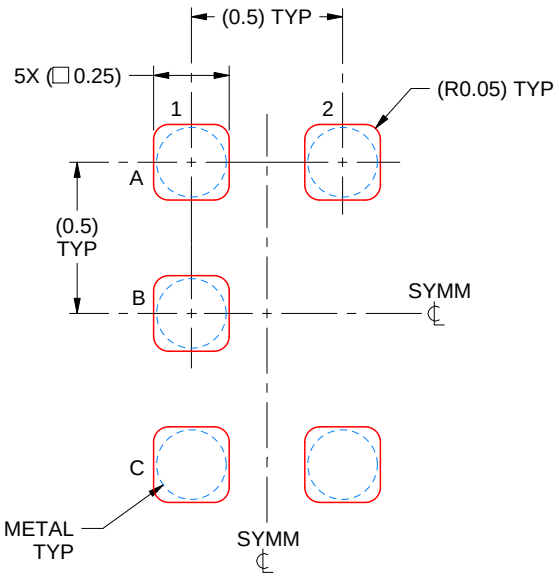


SOLDER MASK DETAILS
NOT TO SCALE

4219492/A 05/2017

NOTES: (continued)

3. Final dimensions may vary due to manufacturing tolerance considerations and also routing constraints. For more information, see Texas Instruments literature number SNVA009 (www.ti.com/lit/snva009).



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:40X

4219492/A 05/2017

NOTES: (continued)

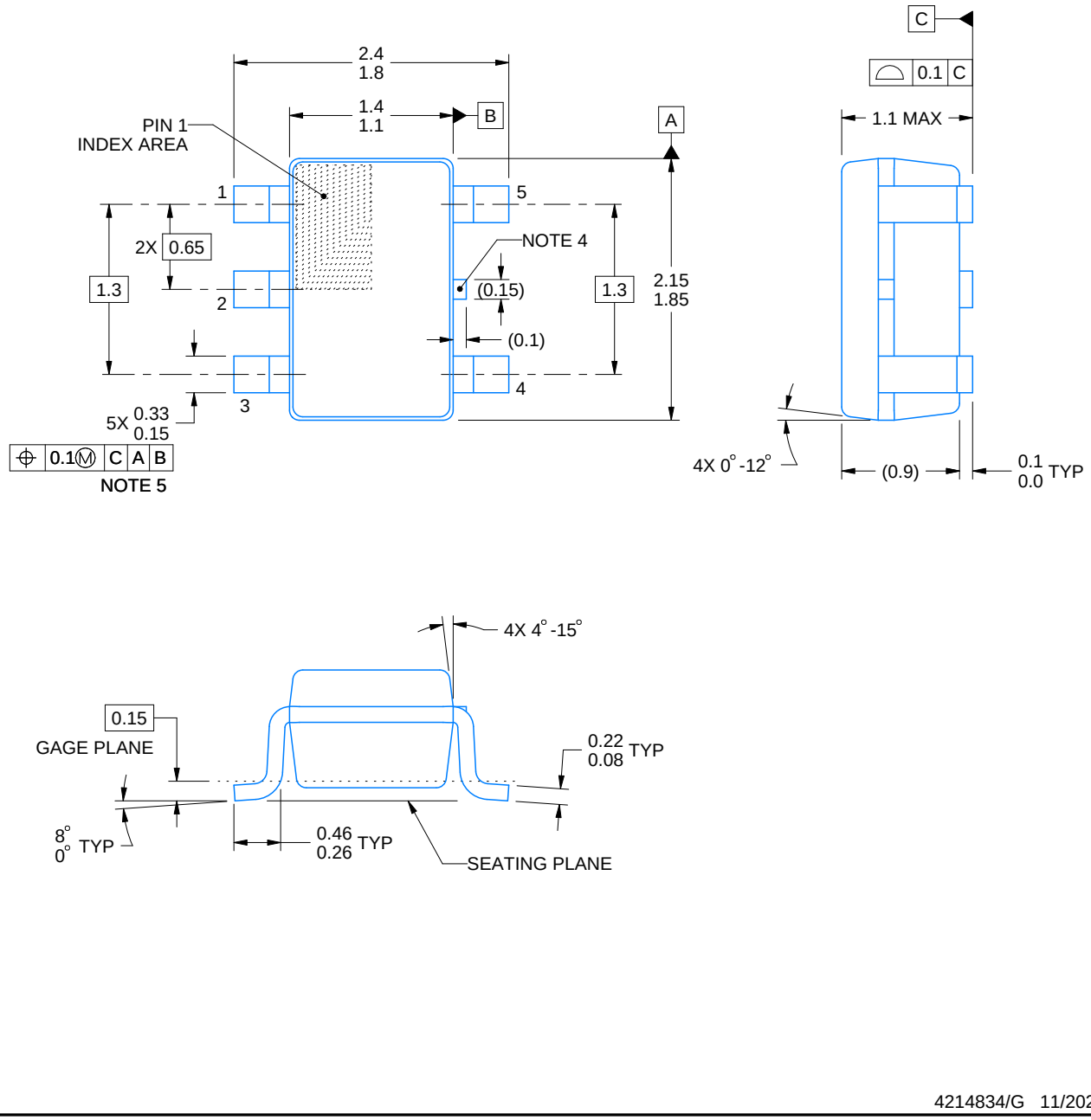
4. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release.

PACKAGE OUTLINE

DCK0005A

SOT - 1.1 max height

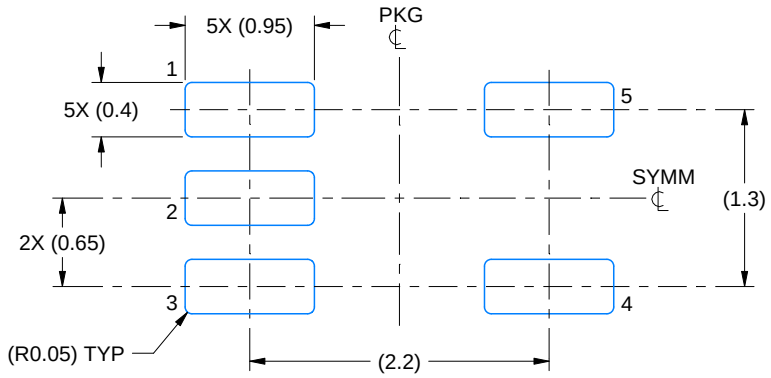
SMALL OUTLINE TRANSISTOR



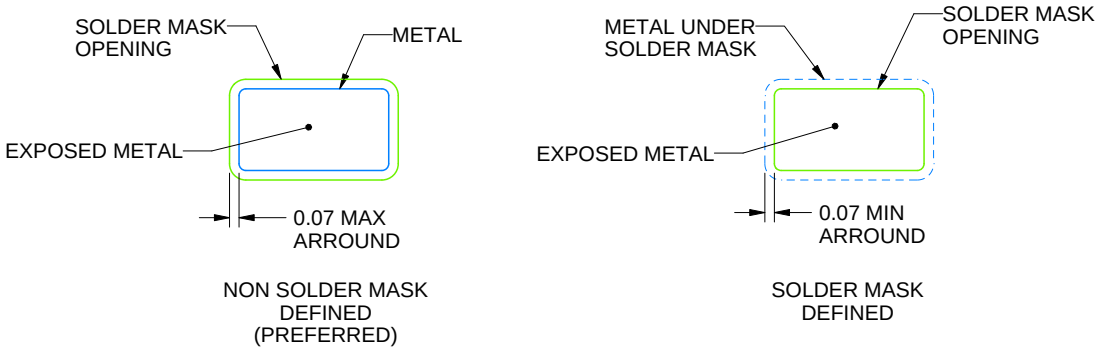
4214834/G 11/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Refernce JEDEC MO-203.
4. Support pin may differ or may not be present.
5. Lead width does not comply with JEDEC.
6. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25mm per side



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

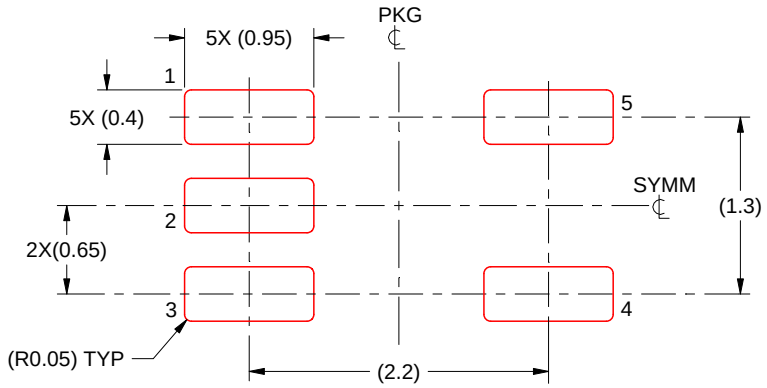


SOLDER MASK DETAILS

4214834/G 11/2024

NOTES: (continued)

7. Publication IPC-7351 may have alternate designs.
8. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 THICK STENCIL
SCALE:18X

4214834/G 11/2024

NOTES: (continued)

9. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
10. Board assembly site may have different recommendations for stencil design.

IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATASHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES “AS IS” AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, regulatory or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you fully indemnify TI and its representatives against any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI’s products are provided subject to TI’s Terms of Sale, TI’s General Quality Guidelines, or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI’s provision of these resources does not expand or otherwise alter TI’s applicable warranties or warranty disclaimers for TI products. Unless TI explicitly designates a product as custom or customer-specified, TI products are standard, catalog, general purpose devices.

TI objects to and rejects any additional or different terms you may propose.

Copyright © 2026, Texas Instruments Incorporated

Last updated 10/2025